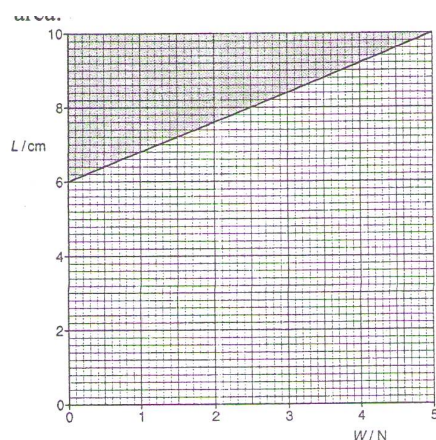


- 6 (a) (i)** The resultant force acting on a body is the rate of change of momentum of the body and acts in the direction of the change in momentum. The unit for the force is newton, which is the force required to accelerate a mass of 1 kg at a rate of 1 m s^{-2} in the direction of the force applied.

(b) (i) The energy stored is represented by:



(ii) Energy stored in spring is work done to stretch it. This can be found by

$$E = \int F dx = \int kx dx = \frac{1}{2} kx^2$$

Alternatively, this is also the triangular area under force-extension graph.

$$E = \frac{1}{2} Fx = \frac{1}{2} kx^2$$

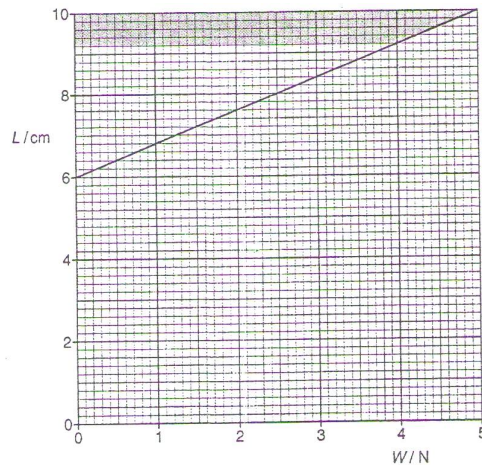
Note: Most answers indicated area under graph without identification of F and W. It was expected that starting point would be a statement that work done is the product of average force and extension.

(c) (i) From the graph, the length of the spring is 9.2 cm when the weight is 4.0 N. Adding another 0.8 cm to this, the total length becomes 10 cm.

(ii) 1. The change in gravitational potential energy

$$mgh = (4.0)(0.8 \times 10^{-2}) = 0.032 \text{ J}$$

2. The change in elastic potential energy is as shown:



$$\begin{aligned}
 & \frac{1}{2}kx_2^2 - \frac{1}{2}kx_1^2 \\
 &= \frac{1}{2}(125)(0.04)^2 - \frac{1}{2}(125)(0.032)^2 \\
 &= 0.036 \text{ J}
 \end{aligned}$$

- (iii) Work done to cause the additional 0.80 cm of extension
 $0.036 - 0.032$
 $= 4.0 \times 10^{-3} \text{ J}$

(d) (i) $4.0 \times 10^{-3} \text{ J}$.

- (ii) 1. Maximum speed occurs when the mass possesses maximum kinetic energy while moving pass its equilibrium position, and its kinetic energy is equal to its total energy:

$$\begin{aligned}
 \frac{1}{2}mv^2 &= 4.0 \times 10^{-3} \text{ J} \\
 \frac{1}{2}\left(\frac{4.0}{9.81}\right)v^2 &= 4.0 \times 10^{-3} \text{ J} \\
 v &= 0.14 \text{ m s}^{-1}
 \end{aligned}$$

2. For spring mass system,

$$\begin{aligned}
 f &= \left(\frac{1}{2\pi}\right)\left(\sqrt{\frac{k}{m}}\right) \\
 &= 2.8 \text{ Hz}
 \end{aligned}$$

- (e) Spring mass would increase effective mass of system increasing period and reducing natural frequency of system.
 Since the spring's force constant is unchanged, the acceleration of the system will decrease according to Newton's second law of motion. Using the defining equation for simple harmonic motion, a smaller acceleration will result in a lower angular frequency and frequency of oscillation.

Note: Must state that spring with mass would increase effective mass of system.

7. (a) A progressive wave is a wave in which the waveform advances and there is a